

Ayan Amanullah Khan - Data Science

Bangalore-India | +91-8792547817 |  [mail:ayyyankhaann3971@gmail.com](mailto:ayyyankhaann3971@gmail.com) |  [ayanamanullahkhan](#) |  [Ayankhan79](#)

SUMMARY

As a **Final-year Data Science student** with a strong foundation in **Machine Learning, Data Analytics, and Python**, and practical working knowledge of **Generative AI, Large Language Models (LLMs), and Agentic AI** concepts. Experienced in applying data analysis and ML techniques through hands-on academic and personal projects, including **predictive modeling, data preprocessing, feature engineering, and model evaluation**. Have built **AI-driven applications and data pipelines** that focus on turning raw data into meaningful insights. Highly interested in developing **practical, scalable data solutions** and continuously learning emerging AI technologies.

EDUCATION

MVJCE, Bangalore	Bangalore
<i>B.E in Computer Science (Data Science) – 8th Sem</i>	<i>Graduation Date: Jun 2026</i>

Sri Venkataramana Pre-University College, Kundapura	Kundapura
<i>PCMS (Mathematics & Statistics)</i>	<i>Graduation Year: 2022</i>

SKILLS & INTERESTS

Programming Skills: Python, HTML, CSS, JavaScript, GO

Database: SQL, PostgreSQL, SQL Server, SQLite, MySQL, MongoDB

Technologies/Frameworks: Django, Flask, Pandas, Open AI, Sklearn, Apache-Pyspark, Tensorflow

Version-Control/Cloud/Tools: Git & Github, Azure, Databricks, MS Excel, PowerBI

Domain- Backend-Development, Data Science, Machine Learning, Large Language Models (LLMs), Extract Transform Load (ETL), Natural Language Processing, Generative AI, Prompt Engineering, Agentic AI

WORK EXPERIENCE

UNLOX AI Solutions Developer Intern (ON-GOING) Assisted in AI solution workflows , including data preparation, ML model training, evaluation, and deployment-oriented understanding. Performed data cleaning, preprocessing, feature engineering, EDA, and data visualization to analyze trends and patterns. Supported machine learning model development for classification and prediction using Python and standard ML libraries . Worked with Generative AI concepts , including LLM usage, prompt engineering, agentic AI basics , and SQL-based data analysis . Gained exposure to backend integration concepts , model evaluation metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC), and documentation/version control practices .	Bangalore
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SOFT SKILLS

CRITICAL THINKING | PROBLEM-SOLVING | EFFECTIVE COMMUNICATION | COLLABORATION AND TEAMWORK | ATTENTION TO DETAIL | INSIGHTS FINDING | LEARNING CURIOSITY

PROJECT EXPERIENCE

ETL Pipeline via github to Azure Cloud

I developed an **end-to-end automated ETL pipeline** on **Microsoft Azure** using a **3-layer Data Lake architecture (Raw, Processed, Curated)**. Implemented **dynamic data ingestion from GitHub** using **Azure Data Factory (ADF)** pipelines. Performed **large-scale data cleaning and transformations** with **Databricks (PySpark)**, applying **schema validation, aggregations, partitioning, and caching** for performance optimization. Modelled and stored **curated datasets in Azure Synapse Analytics** as a **data warehouse** using **schemas and external Tables**. Orchestrated and automated the complete workflow using **Apache Airflow**, ensuring **fault-tolerant, sequential execution**.

Amazon Product Data Analysis (India)

I Conducted an end-to-end analysis of **1,465 Amazon.in product listings** using **Python, Pandas, NumPy, Matplotlib, and Seaborn**. Performed comprehensive data cleaning and preprocessing of real-world pricing, discount, and rating fields, and engineered analytical features such as **price difference, value score (rating × discount percentage), and popularity score (rating count × rating)**. Extracted and structured hierarchical product categories and executed exploratory data analysis to identify **pricing trends, discount patterns, category distribution, and high-performing products**, delivering actionable insights into customer engagement and e-commerce performance.

AI Study Assistant (Offline RAG System) Via Local LLMs

Built an **offline AI-powered PDF question-answering system** using **FastAPI, ChromaDB, and local LLMs via Ollama**, enabling privacy-first semantic search and source-aware responses without cloud dependencies. Implemented a **Retrieval-Augmented Generation (RAG)** pipeline with **PDF ingestion, document chunking, SentenceTransformer embeddings, and vector-based retrieval**, along with a modern full-stack frontend that delivers interactive Q&A and transparent source citations.

AI Stock Trading Agent using Deep Reinforcement Learning

Developed an end-to-end AI-based stock trading system using **Deep Q-Learning (DQN)** to learn optimal trading strategies from historical market data. Designed a custom **trading environment** that simulates real-world buy, sell, and hold actions with portfolio balance tracking and reward calculation. Implemented a **neural network-based Q-function approximator** using PyTorch, along with **experience replay** and **epsilon-greedy exploration** for stable and efficient learning. Trained the agent over multiple episodes on sequential time-series data and evaluated performance through a dedicated testing phase to measure final portfolio balance and profit. The project demonstrates practical application of **reinforcement learning, neural networks, and financial time-series decision-making**.

House Price Prediction Model – Machine Learning Project

Developed an end-to-end machine learning pipeline to predict **median house prices** using the **California Housing dataset**. Performed data preprocessing, feature analysis, and trained multiple regression models including **Linear Regression, Lasso Regression, Random Forest, and XGBoost**. Evaluated model performance using **R² and Mean Absolute Error (MAE)** on training and test sets to assess generalization. Achieved the best results with **XGBoost Regressor**, obtaining a **Test R² of 0.836** and **MAE of 0.31**, significantly outperforming linear models. The project demonstrates strong understanding of **model comparison, bias–variance tradeoff, and non-linear learning methods**.

Customer Churn Prediction Model– Machine Learning Project

Developed an end-to-end **customer churn prediction system** using a telecom dataset with **7,043 records and 21 features**. Performed **data cleaning, feature encoding, and EDA** to identify churn drivers such as **tenure, contract type, payment method, and service usage**, and addressed **class imbalance** using **SMOTE**. Trained and evaluated **Decision Tree, Random Forest, and XGBoost** models with **5-fold cross-validation**, selecting **Random Forest** as the best performer (**84% CV accuracy, 78% test accuracy**). Assessed performance using **confusion matrix, precision, recall, and F1-score**, achieving a **churn recall of 59%**, and proposed improvements including **hyperparameter tuning and stratified cross-validation**.

CERTIFICATIONS AND COURSES

Data Engineering Bootcamp – 5minsEngineering	The Complete AI Coding Course 2025 - Udemy
Complete Machine Learning Course – Udemy	AI Engineer Core Track: LLM Engineering, RAG, QLORA, Agents
Complete Data Science Course 2026 – Udemy	AI Engineer Agentic Track: The Complete Agent & MCP Course
Python Data Structures & Algorithms – Udemy	13+ Power BI Portfolio Projects with DAX and SQL 2026 - Udemy
Mathematics-Basics to Advance for Data Science and GenAI – Udemy	